

International Comparative Studies of Student Assessment

- Korea has been participating international assessments such as Programme for International Student Assessment(PISA), Trends in International Mathematics and Science Study(TIMSS), and The International Computer and Information Literacy Study(ICILS).
- In Korea, PISA, TIMSS and ICILS have been conducted by Korea Institute for Curriculum and Evaluation(KICE) funded by Ministry of Education.



International Comparative Studies of Student Assessment		PISA	TIMSS	ICILS
Target		15 years old	4th graders & 8th graders	8th graders
Research Center	International	OECD	IEA	IEA
	Domestic	KICE	KICE	KICE
Cycle		3 years	4 years	5 years
Assessment Tool	Tests	Reading, Mathematics, Science, Problem Solving	Mathematics, Science	Computer and Information Literacy
	Questionnaires	Student, School, Parents, Teacher*	Student, School, Teacher, Curriculum	Student, Principal, Teacher, ICT-coordinator
Assessment Mode		Paper-based Assessment, Computer-based Assessment	Paper-based Assessment	Computer-based Assessment

* Added to PISA 2015

Recent Schedule

	PISA	TIMSS	ICILS
2010	PISA 2009 Reporting	TIMSS 2011 Field Trial	-
2011	PISA 2012 Field Trial	TIMSS 2011 Main Survey	-
2012	PISA 2012 Main Survey	TIMSS 2011 Reporting	ICILS 2013 Field Trial
2013	PISA 2012 Reporting	Development of TIMSS 2015 Assessment Tools	ICILS 2013 Main Survey
2014	PISA 2015 Field Trial	TIMSS 2015 Field Trial	ICILS 2013 Reporting

What is ICILS?

International Computer and Information Literacy Study(ICILS) is the first administered international comparative study by International Association for the Evaluation of Educational Achievement(IEA) that measures students' Computer and Information Literacy. This study also assesses the computer related background information.

Purpose

ICILS aims to utilize the basic data for establishing the educational policies of participating countries that affect the achievement of educationally coherent information linked to achievement.

Content

ICILS assesses students' preparedness for life in the information age and the ability to use computers to investigate, create and communicate in order to participate effectively at home, at school, in the workplace and in the community.

Assessment Target

ICILS is administered to the eighth graders.

Assessment Cycle

ICILS takes place in a five-year cycle. ICILS 2013, the first assessment, is being prepared.

Assessment Framework

ICILS 2013	Strand	Aspect
Computer and Information Literacy	Collecting and managing information	• Knowing about and understanding computer use • Accessing and evaluating information • Managing information
	Producing and exchanging information	• Transforming information • Creating information • Sharing information • Using information safely and securely
Contextual Questionnaires	Student, Principal, Teacher, ICT-Coordinator	

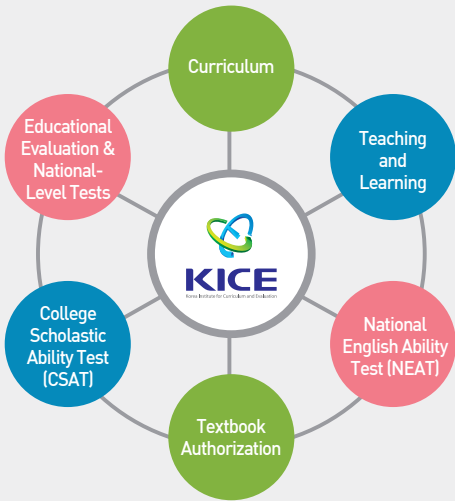
PISA/TIMSS/ICILS

International Comparative Studies of Student Assessment



About KICE

Korea Institute for Curriculum and Evaluation(KICE) is a government-funded educational research institute whose purpose is to contribute to the higher quality of school education, the growth of the nation's public education with research, the development of curricula for elementary and secondary schools, and the implementation of various educational evaluation and tests.



MISSION

Contribution to the higher quality of school education and the national development through research on curricula and evaluation for elementary and secondary schools

VISION

The Research Institute for Curriculum and Evaluation : leadership in the cultivation of creative human resources

TARGET

	Mid-and long-term goals	Strategy
01	A research institute for curriculum and evaluation leading and supporting the educational policy agenda through high quality research	Leading educational policies by increasing the quality of research and dissemination of outcomes
02	A national-level, evaluation-oriented institute obtaining public trust through a stable and advanced assessment	Stabilizing and modernizing national-level evaluation
03	A government-funded research institute maximizing the capability of organization through dynamic and ethical management	Increasing credibility through dynamic and ethical management

<http://www.kice.re.kr>

What is PISA?

Programme for International Student Assessment(PISA) is an international standardized assessment coordinated by the Organisation for Economic Co-operation and Development(OECD) and jointly developed by participating countries.

Purpose

PISA aims to provide participating countries with a chance to investigate students' performance in cognitive domains and educational backgrounds which influence student achievement.

Content

PISA assesses student's literacy in three domains(Reading, Mathematics, and Science) by examining the mastery of processes, the understanding of concepts and the ability to function in various situations within each domain.

Assessment target

PISA is administered to 15-year-old students near to the end of compulsory education.

Assessment cycle

PISA takes place in a three-year cycle. Each of these cycles looks in depth at a major domain; Reading in 2000 and 2009, Mathematics in 2003 and 2012, and Science in 2006 and 2015. PISA 2015, the sixth assessment, is being prepared.

Assessment Framework

PISA 2009	Situation/Context	Aspect/Process/Competency	Text/Content/ Knowledge
Reading	• Personal • Public • Educational • Occupational	• Access and retrieve • Integrate and interpret • Reflect and evaluate	• Medium • Environment • Text format • Text type
Mathematics	• Personal • Public • Educational • Occupational • Intra-Mathematical	• Reproduction • Connections • Reflection	• Quantity • Space and Shape • Change and Relationships • Uncertainty
Science	• Personal • Social • Global	• Identifying scientific issues • Explaining phenomena scientifically • Using scientific evidence	• Knowledge about Science • Knowledge of Science
Contextual Questionnaires	Student, School, Parents		

International Student Performance in PISA 2009

Reading				Mathematics				Science			
Country	Avera-ge score	Rank		Country	Avera-ge score	Rank		Country	Avera-ge score	Rank	
		OECD	Total			OECD	Total			OECD	Total
Shanghai-China	556	-	1	Shanghai-China	600	-	1	Shanghai-China	575	-	1
Korea	539	1~2	2~4	Singapore	562	-	2	Finland	554	1	2~3
Finland	536	1~2	2~4	Hong Kong-China	555	-	3~4	Hong Kong-China	549	-	2~3
Hong Kong-China	533	-	3~4	Korea	546	1~2	3~6	Singapore	542	-	4~6
Singapore	526	-	5~6	Chinese Taipei	543	-	4~7	Japan	539	2~3	4~6
Canada	524	3~4	5~7	Finland	541	1~3	4~7	Korea	538	2~4	4~7
New Zealand	521	3~5	6~9	Liechtenstein	536	-	5~9	New Zealand	532	3~6	6~9
Japan	520	3~6	5~9	Switzerland	534	2~4	6~9	Canada	529	4~7	7~10
Australia	515	5~7	8~10	Japan	529	3~6	8~12	Estonia	528	4~8	7~11
Netherlands	508	5~13	8~16	Canada	527	4~6	9~12	Australia	527	4~8	7~11
Belgium	506	7~10	10~14	Netherlands	526	3~7	8~13	Netherlands	522	4~11	7~16
Norway	503	7~14	10~18	Macao-China	525	-	10~12	Chinese Taipei	520	-	11~15
Estonia	501	8~17	11~21	New Zealand	519	6~8	12~14	Germany	520	7~10	10~15
Switzerland	501	8~17	11~21	Belgium	515	7~11	13~17	Liechtenstein	520	-	10~16
Poland	500	8~17	11~22	Australia	514	7~11	13~17	Switzerland	517	8~12	12~17
Iceland	500	9~16	12~19	Germany	513	8~12	13~17	The United Kingdom	514	9~13	14~19
United States	500	8~20	11~25	Estonia	512	8~11	14~17	Slovenia	512	10~13	16~19
Liechtenstein	499	-	11~23	Iceland	507	11~13	17~19	Macao-China	511	-	16~19
Sweden	497	10~21	13~26	Denmark	503	12~16	18~21	Poland	508	12~16	17~22
Germany	497	11~21	14~26	Slovenia	501	13~15	19~21	Ireland	508	11~17	16~23

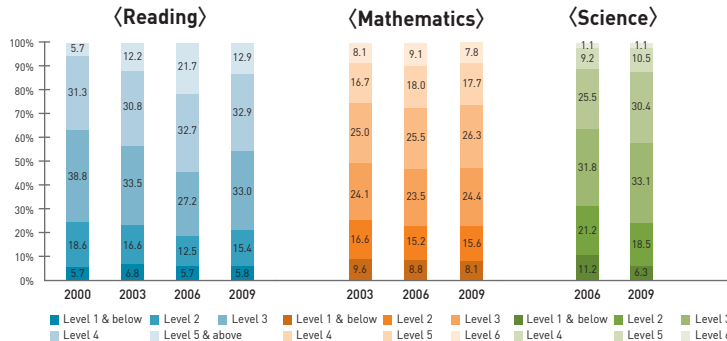
*PISA score is an average of 500 with the standard deviation of 100
*Since PISA 2006, the range of rank for each country has been provided at the 95% confidence level.

Trends of Korean Students' Achievements in PISA

Korea has participated in PISA since the first cycle and has been ranked as one of the top countries continuously.

PISA Cycle	No. of participating countries	Reading		Mathematics		Science	
		Rank	Average score	Rank	Average score	Rank	Average score
2000	43	6	525	3	547	1	552
2003	41	2	534	3	542	4	538
2006	57	1	556	1~4	547	7~13	522
2009	65	2~4	539	3~6	546	4~7	538

*The results of PISA 2012 will be released in the early December 2013.



What is TIMSS?

Trends in International Mathematics and Science Study(TIMSS) is International Association for the Evaluation of Educational Achievement (IEA)'s series of international assessments of student achievement dedicated to improving teaching and learning in Mathematics and Science.

Purpose

TIMSS aims to utilize the basic data for establishing the educational policies of participating countries that affects the achievement of educationally coherent information linked to achievement.

Content

TIMSS assesses Mathematics and Science achievement. In addition, it examines educational context variable through student, teacher, school and curriculum questionnaire.

Assessment Target

TIMSS is administered to the fourth and eighth graders.

Assessment Cycle

TIMSS takes place in a four-year cycle. Since 1995, TIMSS 1995, TIMSS 1999, TIMSS 2003, TIMSS 2007 and TIMSS 2011 have been conducted. TIMSS 2015, the sixth assessment, is being prepared.

Assessment Framework

TIMSS 2011		Content Domains	Cognitive Domains
Mathematics	4th Grade	• Number • Geometric Shapes and Measures • Data Display	• Knowing • Applying • Reasoning
	8th Grade	• Number • Algebra • Geometry • Data and Chance	
Science	4th Grade	• Life Science • Physical Science • Earth Science	
	8th Grade	• Biology • Chemistry • Physics • Earth Science	
Contextual Questionnaires		Student, School, Teacher, Curriculum	

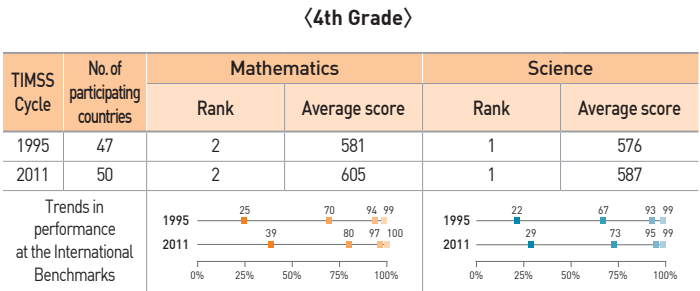
International Student Performance in TIMSS 2011

〈4th Grade〉						〈8th Grade〉					
Mathematics			Science			Mathematics			Science		
Country	Average score	Rank	Country	Average score	Rank	Country	Average score	Rank	Country	Average score	Rank
Singapore	606	1	Korea, Rep. of	587	1	Korea, Rep. of	613	1	Singapore	590	1
Korea, Rep. of	605	2	Singapore	583	2	Singapore	611	2	Chinese Taipei	564	2
Hong Kong SAR	602	3	Finland	570	3	Chinese Taipei	609	3	Korea, Rep. of	560	3
Chinese Taipei	591	4	Japan	559	4	Hong Kong SAR	586	4	Japan	558	4
Japan	585	5	Russian Federation	552	5	Japan	570	5	Finland	552	5
Northern Ireland	562	6	Chinese Taipei	552	5	Russian Federation	539	6	Slovenia	543	6
Belgium (Flemish)	549	7	United States	544	7	Israel	516	7	Russian Federation	542	7
Finland	545	8	Czech Republic	536	8	Finland	514	8	Hong Kong SAR	535	8
England	542	9	Hong Kong SAR	535	9	United States	509	9	England	533	9
Russian Federation	542	9	Hungary	534	10	England	507	10	United States	525	10

*TIMSS score is an average of 500 with the standard deviation of 100

Trends of Korean Students' Achievements in TIMSS

Korea has participated in TIMSS since the first cycle and has been ranked as one of the top countries continuously.



〈8th Grade〉					
TIMSS Cycle	No. of participating countries	Mathematics		Science	
		Rank	Average score	Rank	Average score
1995	40	3	581	4	546
1999	38	2	587	5	549
2003	46	2	589	3	558
2007	50	2	597	4	553
2011	42	1	613	3	560

